computers to have a connection with other devices by using Ethernet cables. The switches and routers may be used to increase the number of systems to be interconnected.

The Ethernet cable is the basic component of the Local Area Network (LAN); here only one popular Ethernet wire is used - RJ45 Ethernet cable. We shall be discussing the different components used inside the Ethernet cable. The components of the Ethernet cable are RJ45 connector, patch cable (UTP cable), plastic covering. The RJ45 connector is made up of plastic cubes connected at the both the ends. With this connector we will connect one end at the computer and other end at the LAN port. The patch cables has eight small wires inside. The plastic covering is used to provide protection to the wires when it bends or get twisted while connecting to the system or to the LAN port.

Ethernet cabling is the process of connecting the computers with other devices using Ethernet cables. The four main components used in the Ethernet cabling components are

1. Patch Cable (Twisted pair)
2. RJ45 Connector
3. Ethernet Ports
4. Crimping Tool

13.3.1 Patch Cable (Twisted Pair)

These Cables are generally made up of 8 wires in different colors. Four of them are solid colours, and the others are striped. The eight colors are white green, green, white orange, blue, white blue, orange, white brown and brown. The following figure 13.8 shows the patch cable.



Figure 13.8 Patch cable (Twisted pair)

Ethernet cables are normally manufactured in several industrial standards such as Cat 3, Cat 5, Cat 6, Cat 6e and cat 7. "Cat" simply stands for "Category," and the following number indicates the version. Latest version denotes faster and higher frequencies, measured in Mhz. Increasing the size of the cable also lead to slower transmission speed.

The cables together with male connectors (RJ45) on each end are commonly referred as Ethernet cables. It is also called as RJ45 cables, since Ethernet cable uses RJ45 connectors

A crossover Ethernet cable is specially designed for making a connection between two computers. Generally, the Ethernet cables are designed to make a connection between a computer and a router or switch.

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